

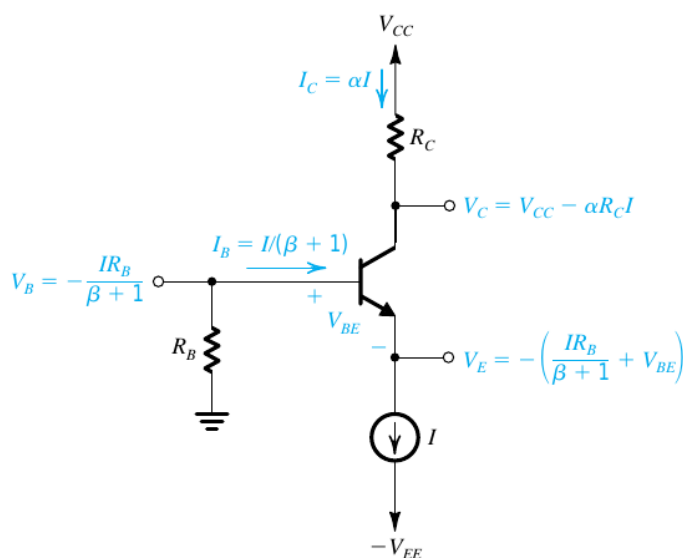
## Lecture 15

### EE-215 Electronic Devices and Circuits

Asst Prof Muhammad Anis Ch

## Discrete-Circuit BJT Amplifiers

- Thus any of the above biasing methods can be utilized to implement a BJT amplifier
  - A BJT amplifier can be a
    - Common Emitter Amplifier
      - Common Base Amplifier
      - or an Emitter Follower
    - here we will utilize the constant-current biasing scheme,
      - to implement the three basic configurations of BJT amplifiers

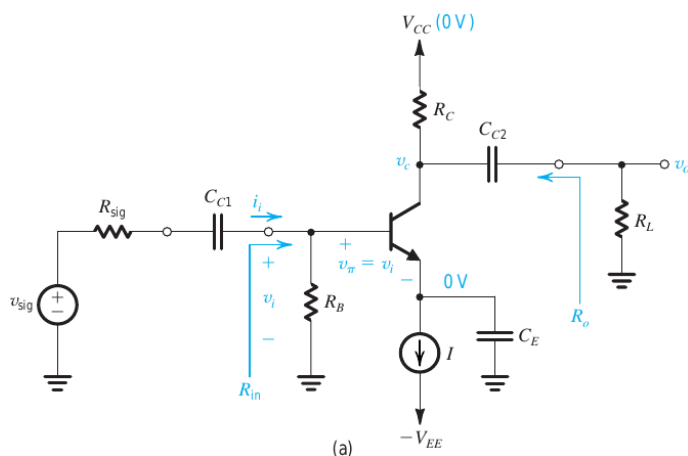


**Figure 6.64** Basic structure of the circuit used to realize single-stage, discrete-circuit BJT amplifier configurations.

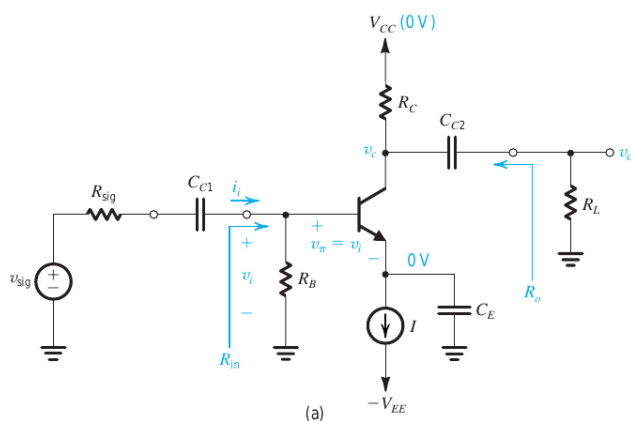
## The Common-Emitter (CE) Amplifier

- A common-emitter amplifier realized using the constant-current biasing is shown here
  - here a signal ground (or ac ground) at the emitter is
    - realized by connecting a capacitor  $C_E$  between the emitter and ground
    - the value of  $C_E$  should be large enough so as to provide a very low impedance at the desired

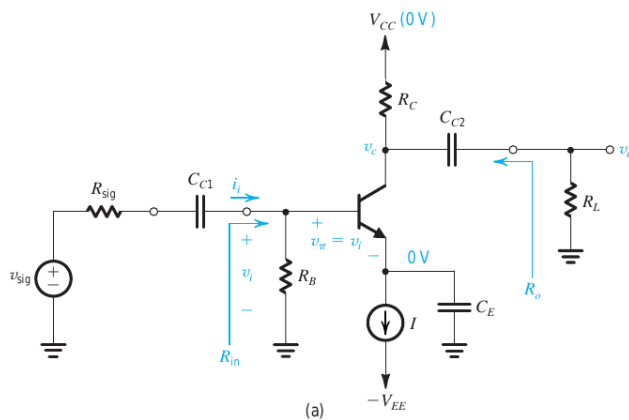
frequencies (i.e.  $\frac{1}{j\omega C_E}$  should be very small)



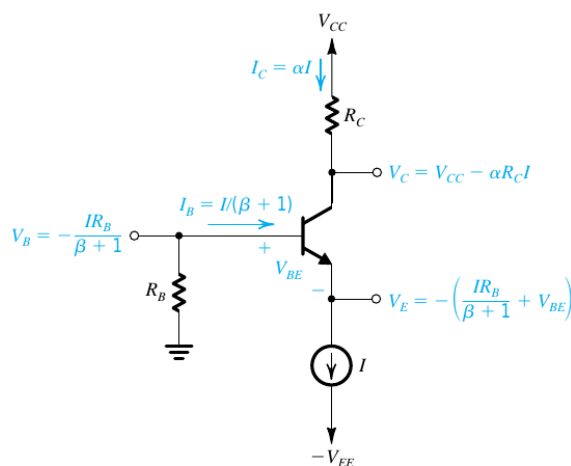
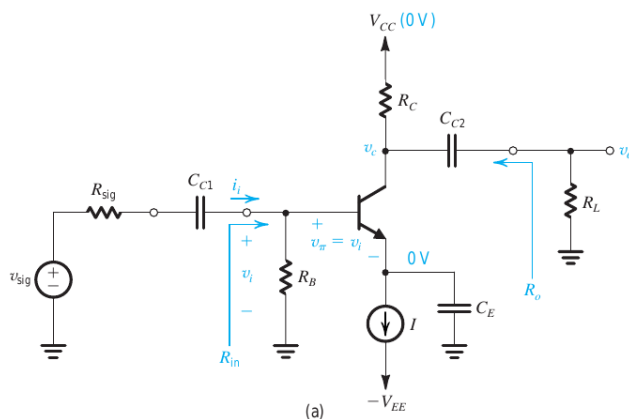
- this capacitor  $C_E$  thus acts as a short circuit for the ac signal frequencies
  - the signal current passes through  $C_E$  to ground and thus bypass the output resistance of current source  $I$
  - $C_E$  is therefore called a bypass capacitor
- $C_{C1}$  is used to couple the ac signal source to the base of BJT
  - this capacitor blocks dc and acts as a short circuit at ac and is called a coupling capacitor



- $C_{C2}$  is also a coupling capacitor which is utilized to connect the load to the collector terminal
  - As a consequence of the coupling and bypass capacitors,
    - the dc equivalent circuit stays the same as the basic constant-current source biasing circuit

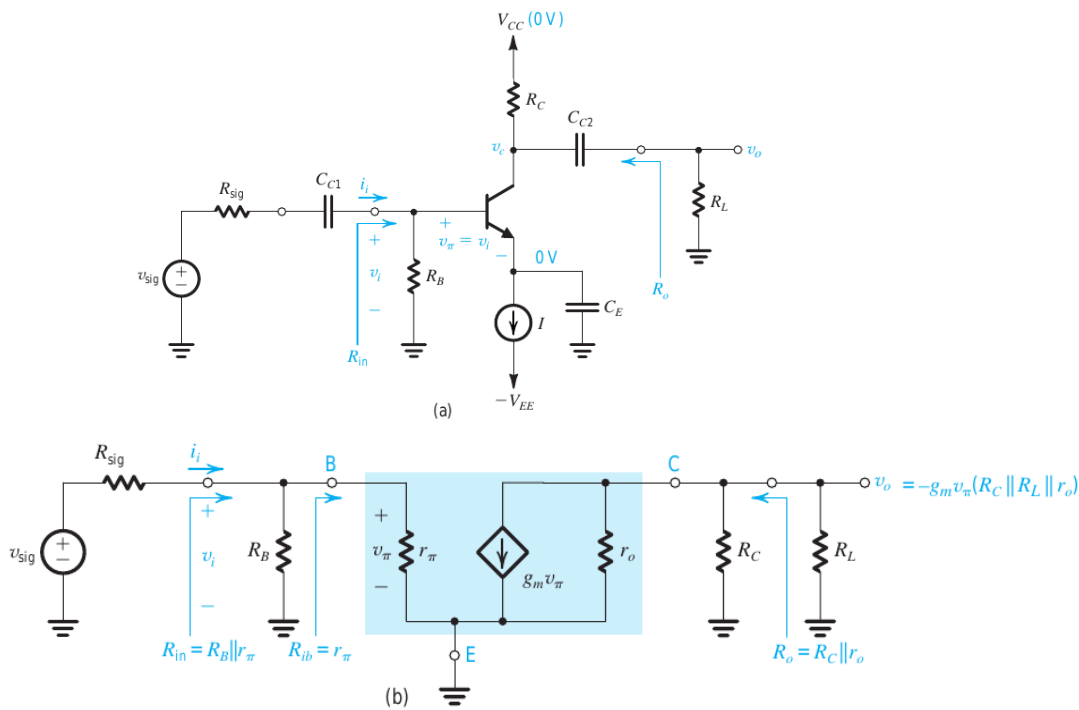


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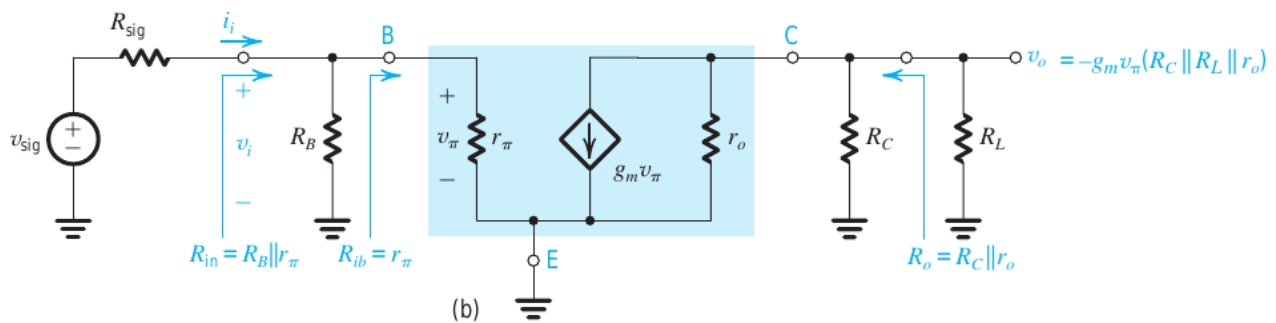
**Figure 6.64** Basic structure of the circuit used to realize single-stage, discrete-circuit BJT amplifier configurations.

- Now to determine the terminal characteristics of the CE amplifier (i.e.  $R_{in}$ ,  $R_o$ ,  $A_v$ ,  $G_v$ ),
  - we replace the CE amplifier by its equivalent small-signal circuit
  - i.e. we replace the BJT by its small-signal model
    - suppress the dc sources
    - replace capacitors by short-circuits (if  $\frac{1}{j\omega C} \rightarrow 0$ )



- from the fig

- $\Rightarrow$  input resistance  $= R_{in} = \frac{v_i}{i_i} = R_B \parallel r_\pi$
- Now the output resistance  $R_o$  is the resistance seen looking into the output terminal with  $v_i$  set to zero
  - $v_i = 0 \Rightarrow v_\pi = 0 (\because v_i = v_\pi) \Rightarrow g_m v_\pi = 0$
  - $\Rightarrow R_o = R_C \parallel r_o$



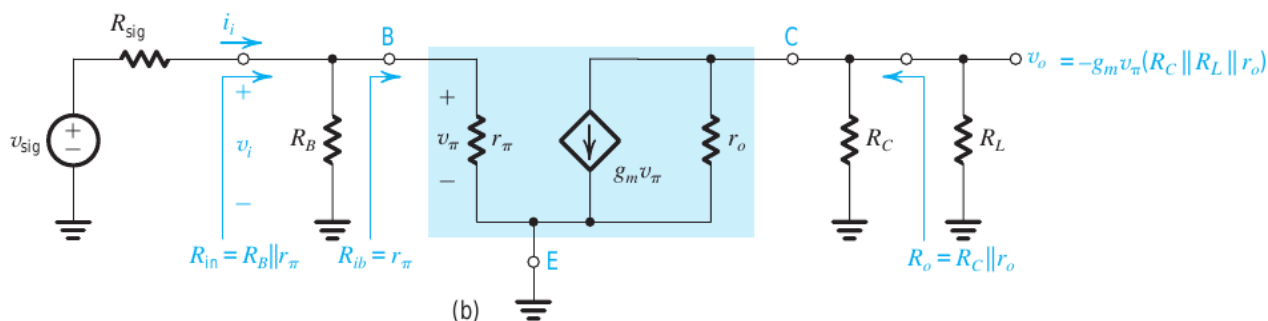
- to determine  $A_{vo}$

- by KCL  $\Rightarrow g_m v_\pi + \frac{v_o}{r_o \parallel R_C} = 0$ 
  - $v_o = -g_m v_\pi (r_o \parallel R_C) \Rightarrow \frac{v_o}{v_\pi} = -g_m (r_o \parallel R_C)$
  - $\frac{v_o}{v_\pi} = -g_m (r_o \parallel R_C)$
  - here  $v_i = v_\pi \Rightarrow$ 

$$A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty} = \left. \frac{v_o}{v_\pi} \right|_{R_L = \infty} = -g_m (r_o \parallel R_C)$$

- thus for CE amplifier

$$\blacksquare R_{in} = R_B \parallel r_{\pi}, R_o = R_C \parallel r_o, A_{vo} = -g_m(r_o \parallel R_C)$$



- now to determine  $A_v$ , the load resistor  $R_L$  is connected

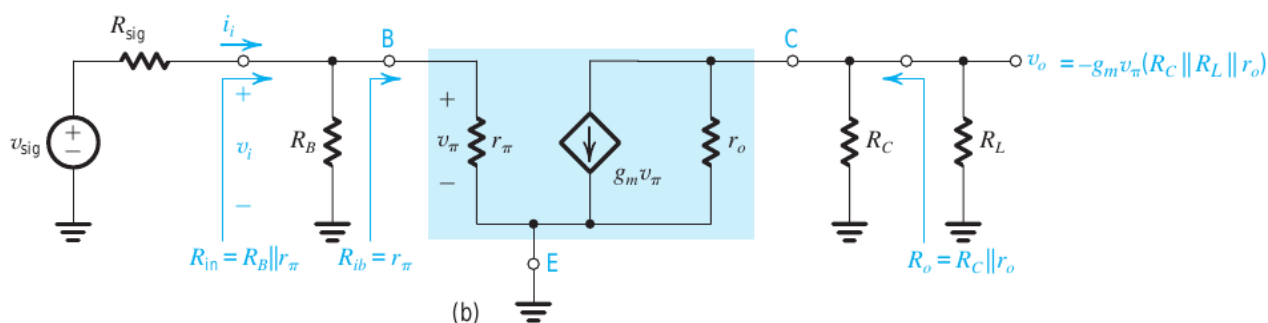
$$\circ \text{ i.e. } A_v = \frac{v_o}{v_i}$$

- by KCL

$$\blacksquare \frac{v_o}{(R_C \parallel r_o \parallel R_L)} = -g_m v_{\pi}$$

$$\blacksquare v_o = (-g_m v_{\pi})(R_C \parallel r_o \parallel R_L)$$

$$\blacksquare \Rightarrow A_v = \frac{v_o}{v_i} = \frac{v_o}{v_{\pi}} = -g_m(R_C \parallel r_o \parallel R_L)$$



- now the overall voltage gain  $G_v$  can be determined as follows

$$\circ G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$$

$$\circ \text{ As } A_v = \frac{v_o}{v_i} = \frac{v_o}{v_{\pi}} = -g_m(R_C \parallel r_o \parallel R_L) \text{ and by voltage divider}$$

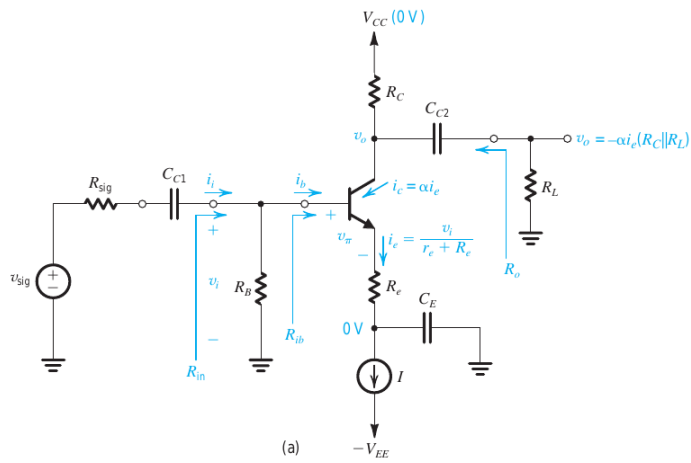
$$\text{rule, } v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$$

$$\blacksquare \text{ As } R_{in} = R_B \parallel r_{\pi} \Rightarrow \frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = \frac{R_B \parallel r_{\pi}}{R_B \parallel r_{\pi} + R_{sig}}$$

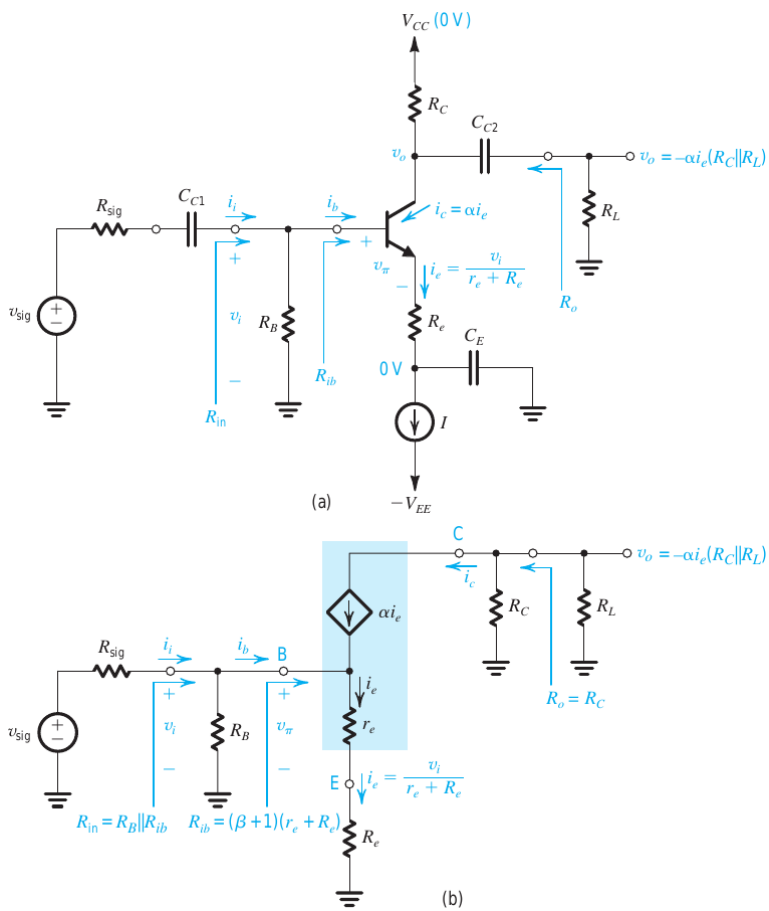
$$\blacksquare \Rightarrow G_v = \frac{v_i}{v_{sig}} A_v = \frac{R_B \parallel r_{\pi}}{R_B \parallel r_{\pi} + R_{sig}} \left\{ -g_m(R_C \parallel r_o \parallel R_L) \right\}$$

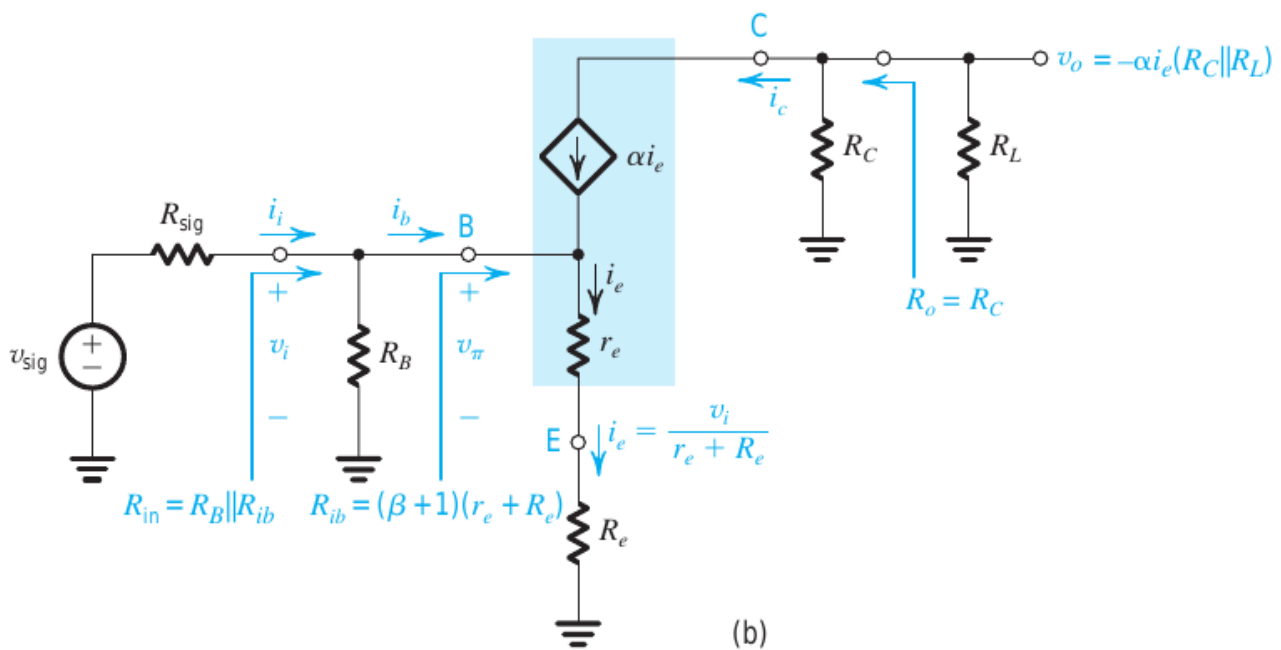
## The Common-Emitter Amplifier with an Emitter Resistance

- the Common-emitter amplifier with an emitter resistance, utilizing the constant-current biasing scheme is shown in figure



- small signal equivalent circuit for fig a, can be given as

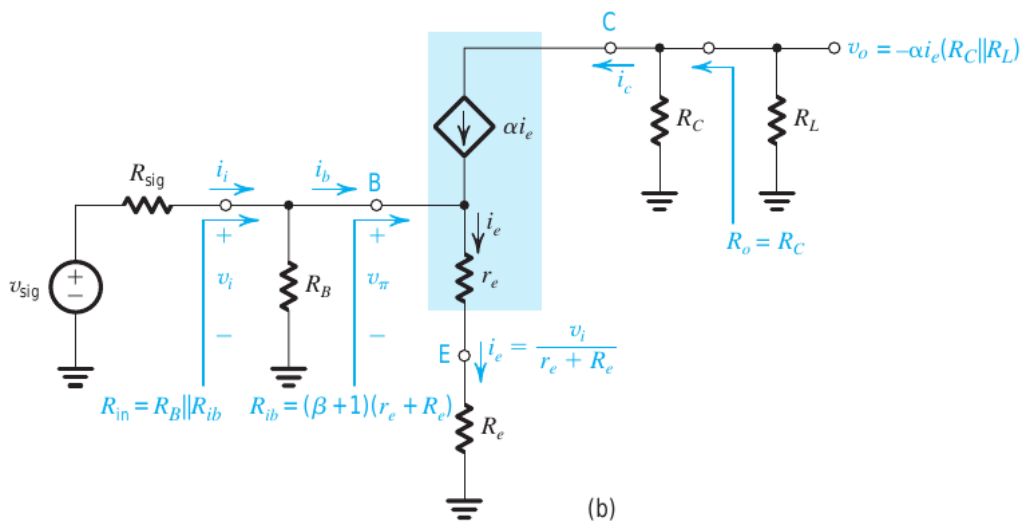




- to determine the amplifier input resistance,  $R_{in}$

$$\circ \Rightarrow R_{in} = \frac{v_i}{i_i} \text{ where } i_i = i_{R_B} + i_b = \frac{v_i}{R_B} + i_b$$

$$\blacksquare \text{ As } i_b = \frac{i_e}{\beta + 1} \Rightarrow i_i = \frac{v_i}{R_B} + \frac{i_e}{\beta + 1}$$



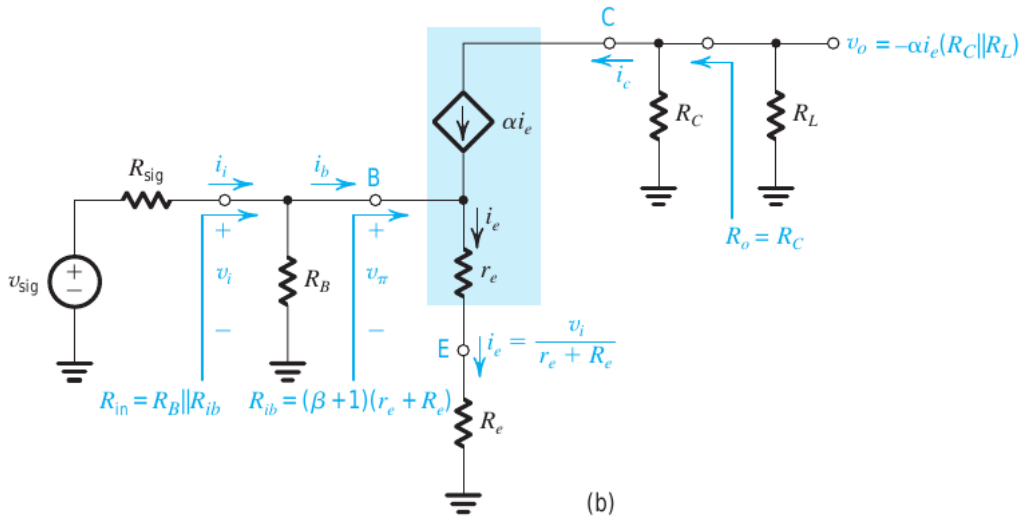
- As  $v_i$  appears across  $r_e + R_e \Rightarrow$  by ohms law

$$\circ i_e = \frac{v_i}{r_e + R_e}$$

$$\blacksquare \Rightarrow i_i = \frac{v_i}{R_B} + \frac{i_e}{\beta + 1} = \frac{v_i}{R_B} + \frac{v_i}{(\beta + 1)(r_e + R_e)}$$

$$\blacksquare \text{ or } i_i = v_i \left( \frac{1}{R_B} + \frac{1}{(\beta + 1)(r_e + R_e)} \right)$$

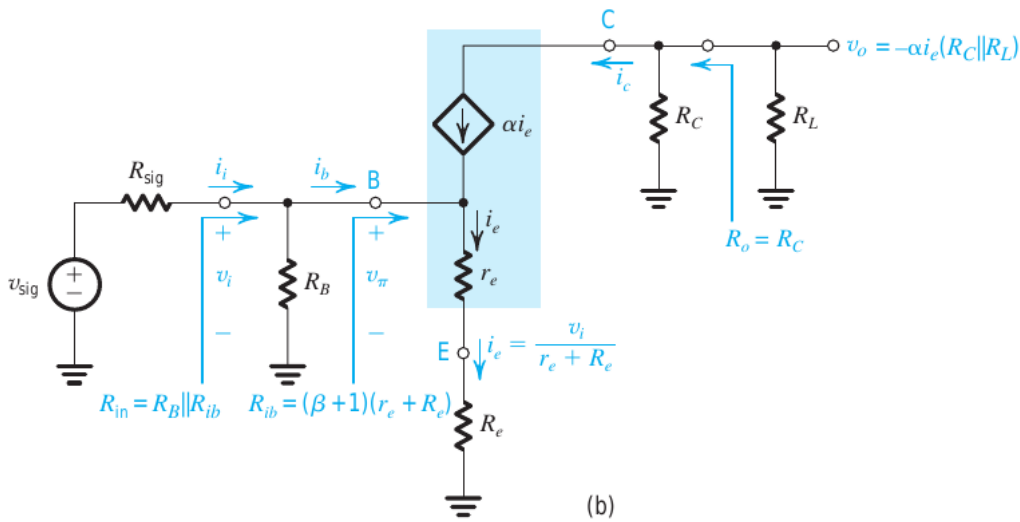
$$\blacksquare \frac{i_i}{v_i} = \frac{1}{R_B} + \frac{1}{(\beta + 1)(r_e + R_e)}$$



- $\frac{i_i}{v_i} = \frac{1}{R_B} + \frac{1}{(\beta+1)(r_e + R_e)}$

$$\circ R_{in} = \frac{v_i}{i_i} = \left( \frac{1}{R_B} + \frac{1}{(\beta+1)(r_e + R_e)} \right)^{-1} = R_B \parallel \left\{ (\beta+1)(r_e + R_e) \right\}$$

- the input resistance looking into the base terminal is  $(\beta + 1)$  times the resistance in the emitter (known as resistance-reflection rule)

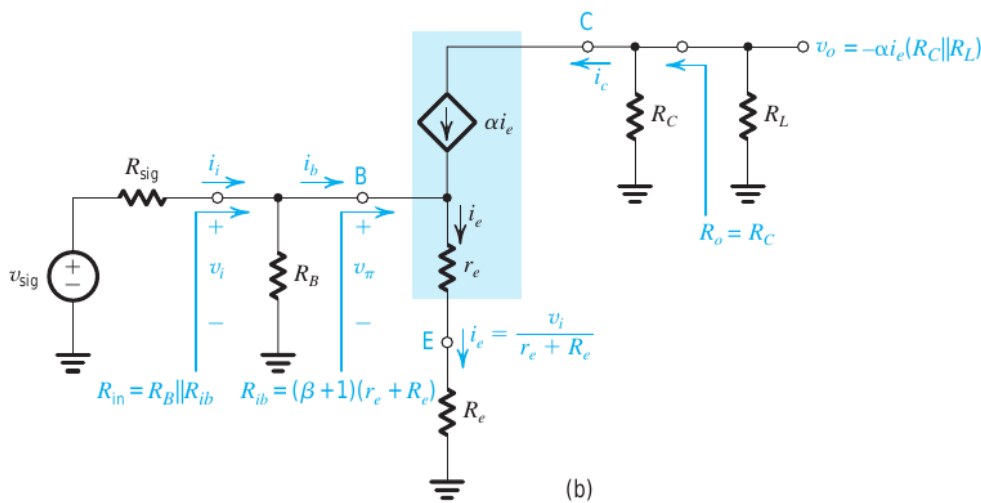


- the output resistance  $R_o$  can be determined by placing a test source at the collector

- and by setting  $v_i = 0 \Rightarrow i_e = \frac{v_i}{r_e + R_e} = 0 \Rightarrow \alpha i_e = 0$

- $\alpha i_e = 0 \Rightarrow R_o = R_C$





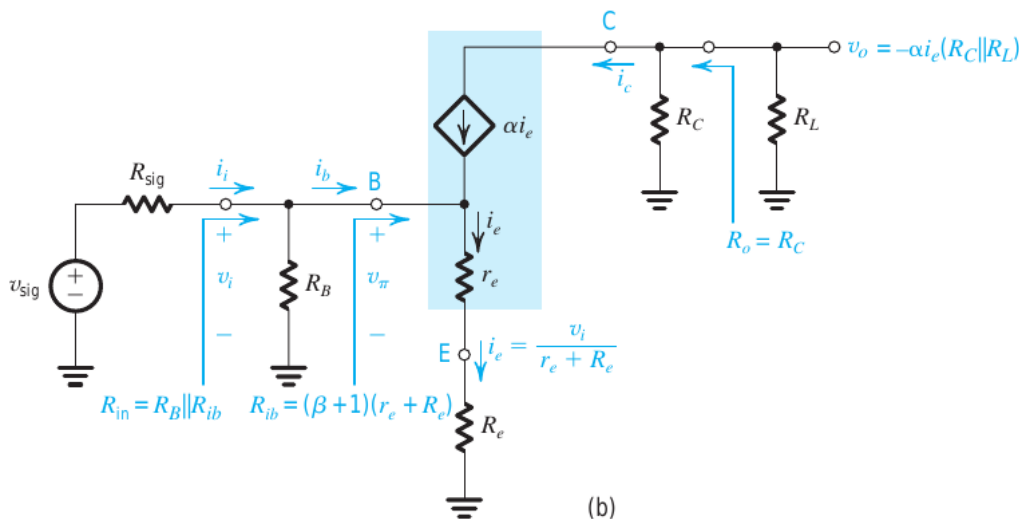
- Now to determine voltage gain  $A_{vo}$

- KCL at C  $\Rightarrow \frac{v_o}{R_C} + \alpha i_e = 0$  or  $\frac{v_o}{R_C} = -\alpha i_e$

- $\Rightarrow v_o = -\alpha i_e R_C$

- by ohm's law  $i_e = \frac{v_i}{r_e + R_e} \Rightarrow v_o = -\alpha R_C \frac{v_i}{r_e + R_e}$

- or  $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty} = -\alpha \frac{R_C}{r_e + R_e}$



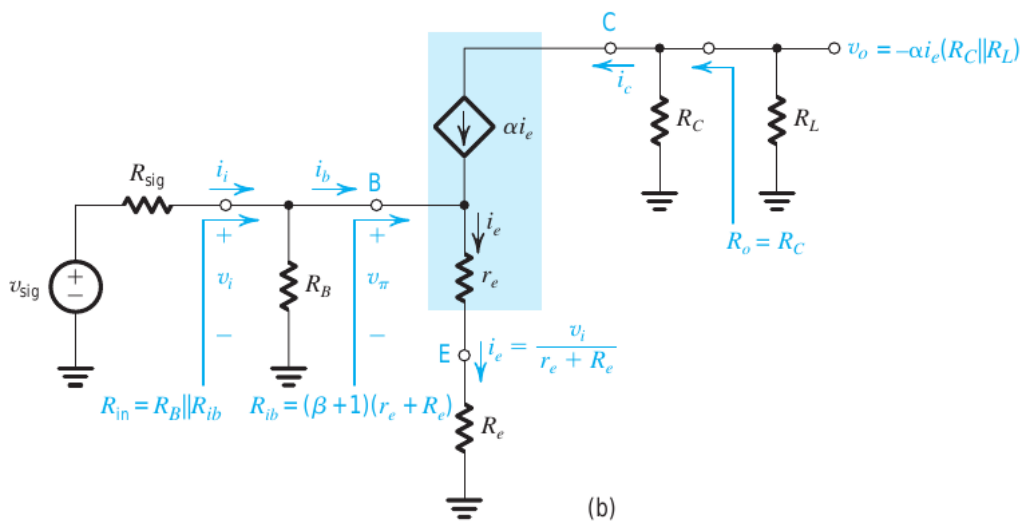
- $A_{vo} = -\alpha \frac{R_C}{r_e + R_e}$

- now when  $R_L$  is connected at the output,  $A_v$  can be given as

- As  $R_L$  appears in parallel with  $R_C$

- thus by replacing  $R_C$  by  $R_C \parallel R_L$  in expression of  $A_{vo}$  gives  $A_v$

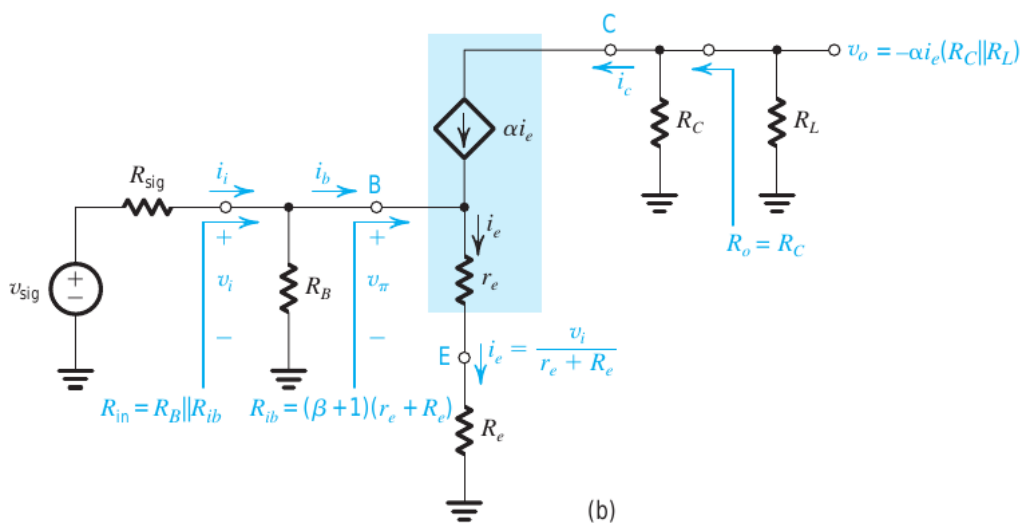
- as  $\Rightarrow A_v = -\alpha \frac{R_C \parallel R_L}{r_e + R_e}$



- Now the overall gain  $G_v$  can be given as

$$\circ G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v$$

$$\blacksquare \quad G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} \left( -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e} \right)$$



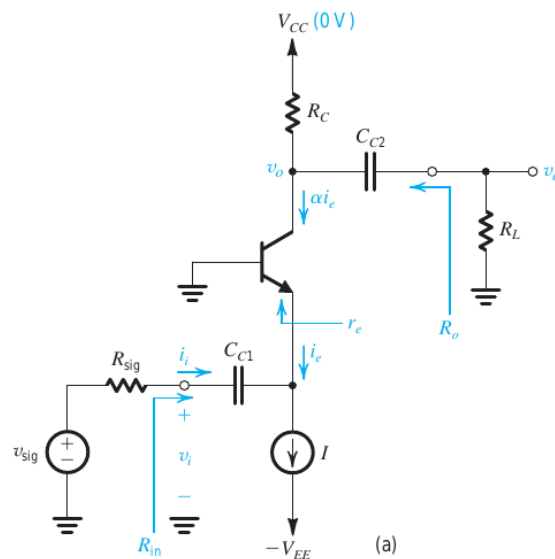
- $G_v = \frac{R_{in}}{R_{in} + R_{sig}} \left( -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e} \right)$

- As  $R_{in} = R_B \parallel \{(\beta + 1)(r_e + R_e)\}$

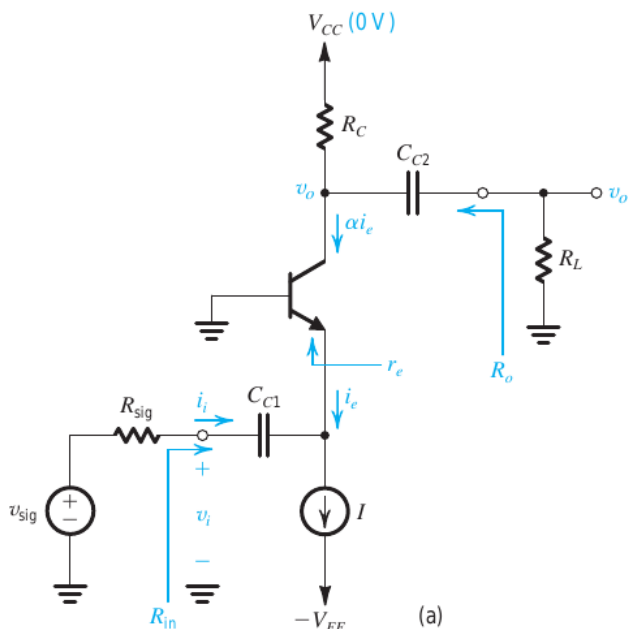
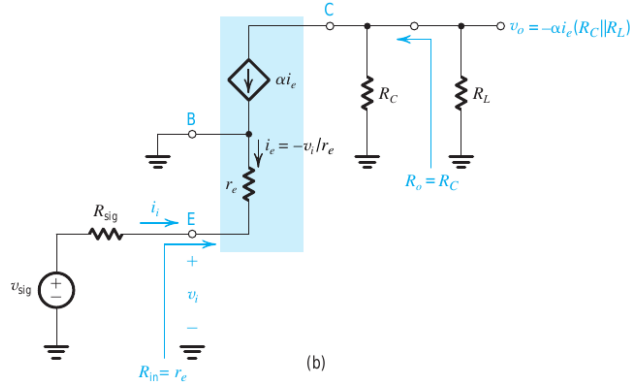
$$0 \Rightarrow G_v = \frac{R_B \parallel \{(\beta+1)(r_e + R_e)\}}{R_B \parallel \{(\beta+1)(r_e + R_e)\} + R_{sig}} \left( -\frac{\alpha(R_C \parallel R_L)}{r_e + R_e} \right)$$

## The Common-Base (CB) Amplifier

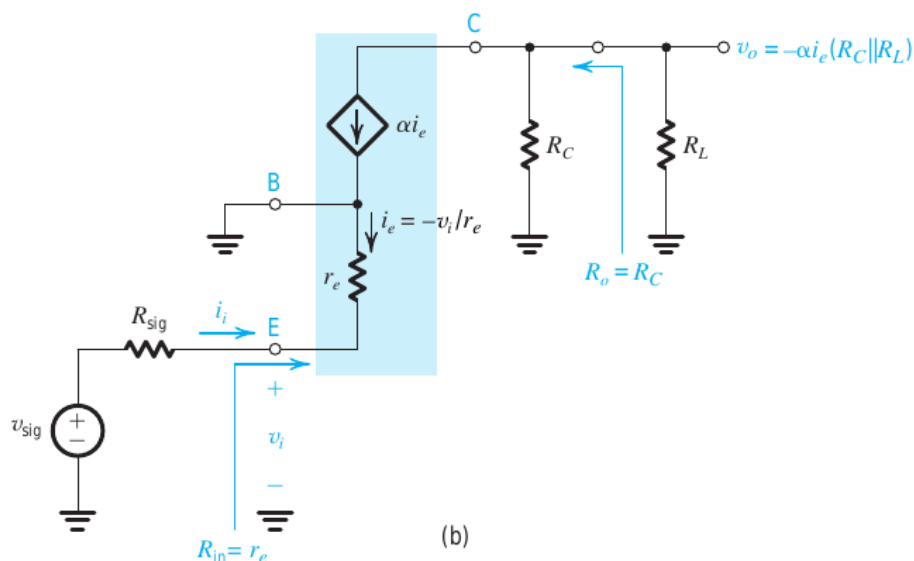
- here both the dc and ac voltages at the base are to be zero,
    - thus the base is directly connected to the ground
- 7.2cm



- the small-signal equivalent circuit can be given as

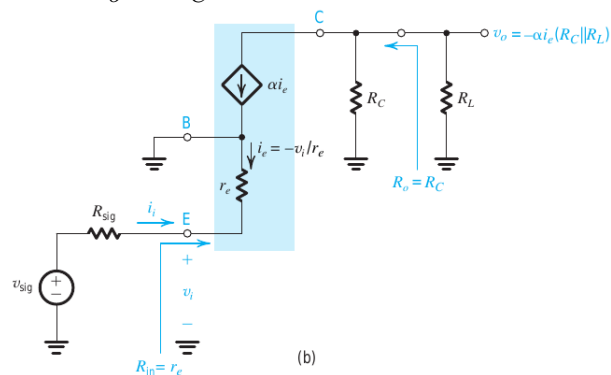


- the input resistance  $R_{in}$  can be given as  $R_{in} = \frac{v_i}{i_i} = \frac{-v_{be}}{-i_e} = \frac{v_{be}}{i_e} = r_e$ 
  - as  $R_{in}$  is the resistance looking into the emitter and the base is grounded



- The output resistance of the CB amplifier is determined by applying a test source at the output and setting  $v_i = 0$

$$\begin{aligned} \circ v_i = 0 &\Rightarrow i_e = \frac{-v_i}{r_e} = 0 \Rightarrow \alpha i_e = 0 \\ \circ &\Rightarrow R_o = R_C \end{aligned}$$



- Now to determine the voltage gain  $A_{vo}$

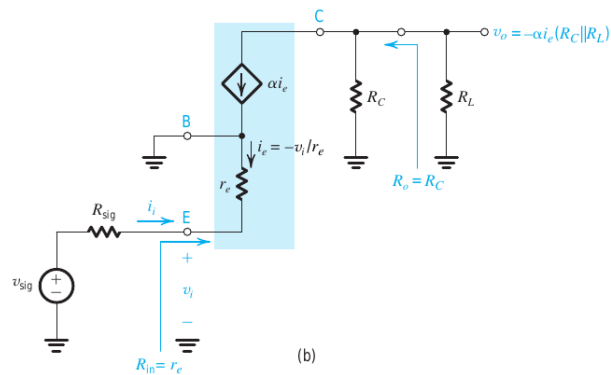
- by KCL at C

$$\blacksquare \frac{v_o}{R_C} + \alpha i_e = 0 \Rightarrow v_o = -\alpha R_C i_e$$

$$\blacksquare v_o = -\alpha R_C i_e$$

- where  $i_e$  is the emitter current i.e.

$$i_e = \frac{0 - v_i}{r_e} = \frac{-v_i}{r_e}$$

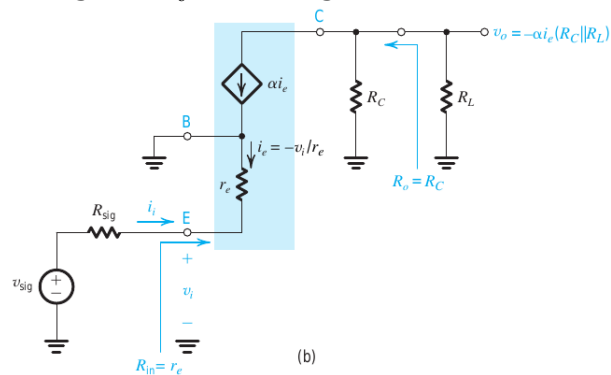


$$\circ \Rightarrow v_o = -\alpha R_C i_e = -\alpha R_C \left( \frac{-v_i}{r_e} \right) = \frac{\alpha R_C v_i}{r_e}$$

$$\circ \Rightarrow A_{vo} = \frac{v_o}{v_i} = \alpha \frac{R_C}{r_e} = g_m R_C \because r_e = \frac{\alpha}{g_m}$$

- As  $A_{vo} = g_m R_C$

- the gain  $A_v$  can be given as



$$\circ A_v = \frac{v_o}{v_i} = g_m (R_C \parallel R_L)$$

$$\blacksquare \because A_{vo} = g_m R_C \text{ and as } R_L \text{ is in parallel with } R_C$$

- To determine the overall gain  $G_v$

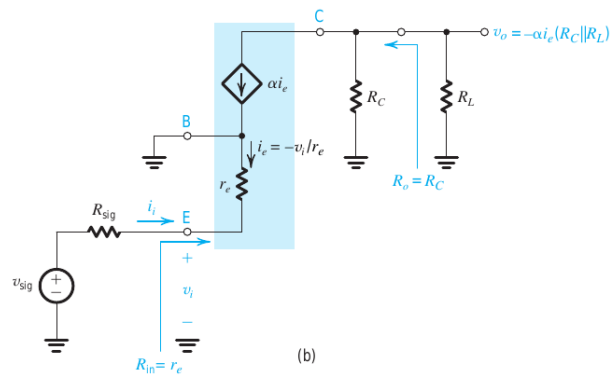
- first by voltage divider

$$\blacksquare v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig} = \frac{r_e}{r_e + R_{sig}} v_{sig} \because R_{in} = r_e$$

$$\circ G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i}$$

$$\blacksquare G_v = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v = \frac{r_e}{r_e + R_{sig}} A_v$$

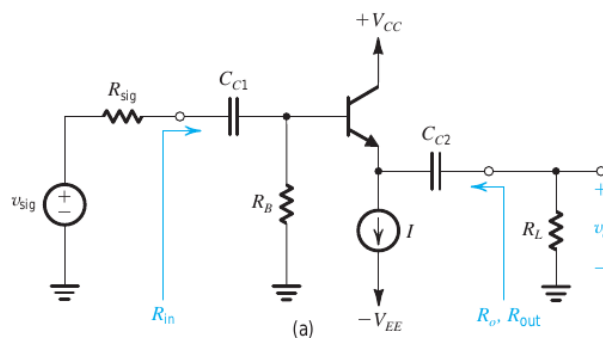
$$\blacksquare \because A_v = g_m (R_C \parallel R_L)$$



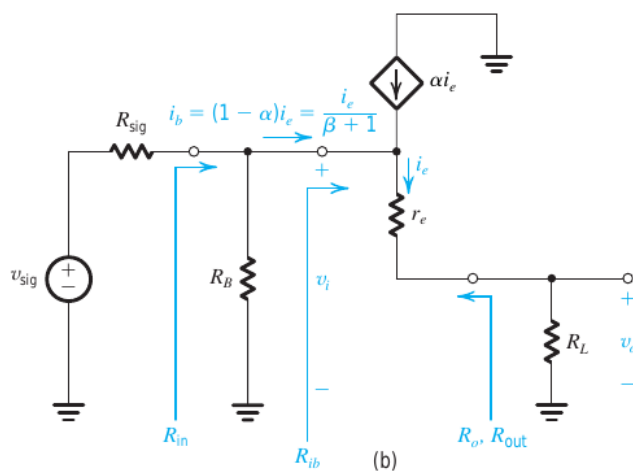
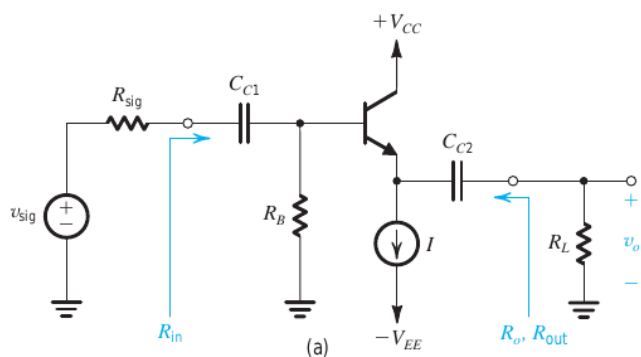
- $G_v = \frac{r_e}{r_e + R_{sig}} g_m (R_C \parallel R_L)$
- $G_v = \frac{r_e}{r_e + R_{sig}} g_m (R_C \parallel R_L)$ 
  - As  $r_e = \frac{\alpha}{g_m} \Rightarrow r_e g_m = \alpha$ 
    - $G_v = \frac{\alpha}{r_e + R_{sig}} (R_C \parallel R_L)$
  - $G_v = \frac{\alpha}{r_e + R_{sig}} (R_C \parallel R_L) = \alpha \frac{(R_C \parallel R_L)}{r_e + R_{sig}}$

## The Emitter Follower

- In this common collector amplifier,
  - As the collector is to function as a signal ground, there is no need for resistor  $R_C$
  - therefore  $R_C$  has been eliminated
    - the input signal is applied via  $C_{C1}$  to the base
    - the output signal is taken at the emitter and is coupled to the load  $R_L$  via  $C_{C2}$



- Small signal equivalent circuit for this amplifier can be obtained
  - by replacing BJT with its T-model,
    - by suppressing dc sources
    - and by replacing capacitors by short circuits (as  $\frac{1}{j\omega C}$  is very small at the frequency of interest)



- from figure, the input resistance  $R_{in}$  can be given as

- $R_{in} = \frac{v_i}{i_i}$  where  $i_i = i_{R_B} + i_b = \frac{v_i}{R_B} + i_b$

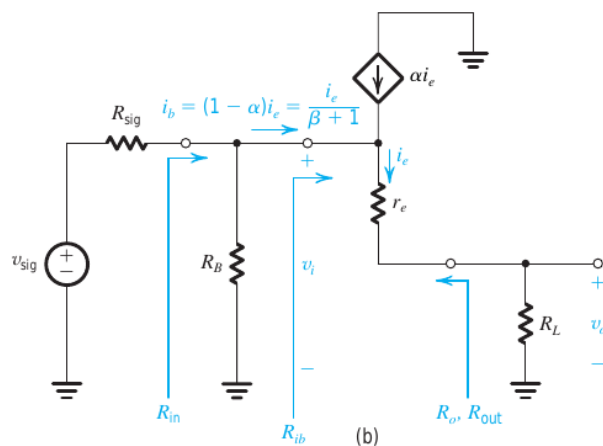
- using KCL  $i_b + \alpha i_e = i_e \Rightarrow i_b = i_e(1 - \alpha)$

- $i_b = i_e \left(1 - \frac{\beta}{\beta + 1}\right) = i_e \left(\frac{\beta + 1 - \beta}{\beta + 1}\right) = \frac{i_e}{\beta + 1}$

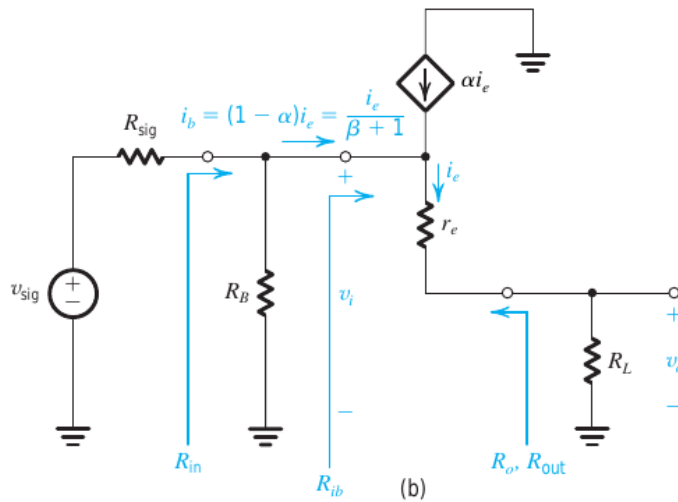
- $\Rightarrow i_i = \frac{v_i}{R_B} + i_b = \frac{v_i}{R_B} + \frac{i_e}{\beta + 1}$

- As by ohms law  $i_e = \frac{v_i}{r_e + R_L} \Rightarrow i_i = \frac{v_i}{R_B} + \frac{v_i}{(\beta + 1)(r_e + R_L)}$

- or  $\frac{i_i}{v_i} = \frac{1}{R_B} + \frac{1}{(\beta + 1)(r_e + R_L)}$



- $\frac{i_i}{v_i} = \frac{1}{R_B} + \frac{1}{(\beta + 1)(r_e + R_L)}$

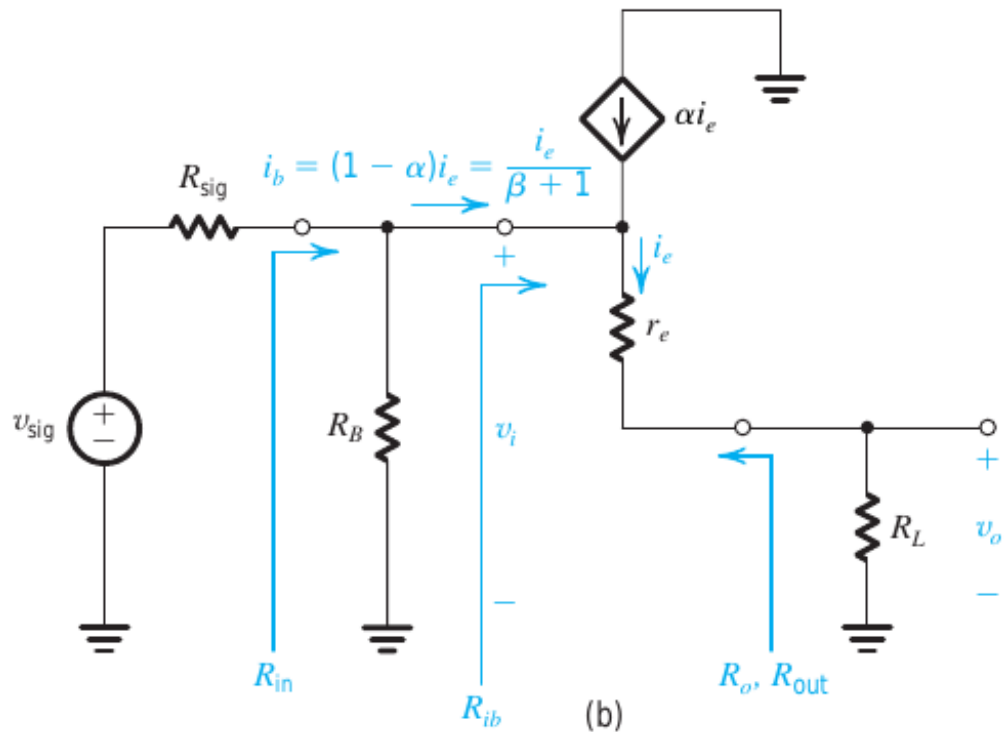


$$\circ \Rightarrow R_{in} = \frac{v_i}{i_i} = \left( \frac{1}{R_B} + \frac{1}{(\beta+1)(r_e + R_L)} \right)^{-1}$$

$$\circ \text{ or } R_{in} = R_B \parallel \{(\beta+1)(r_e + R_L)\}$$

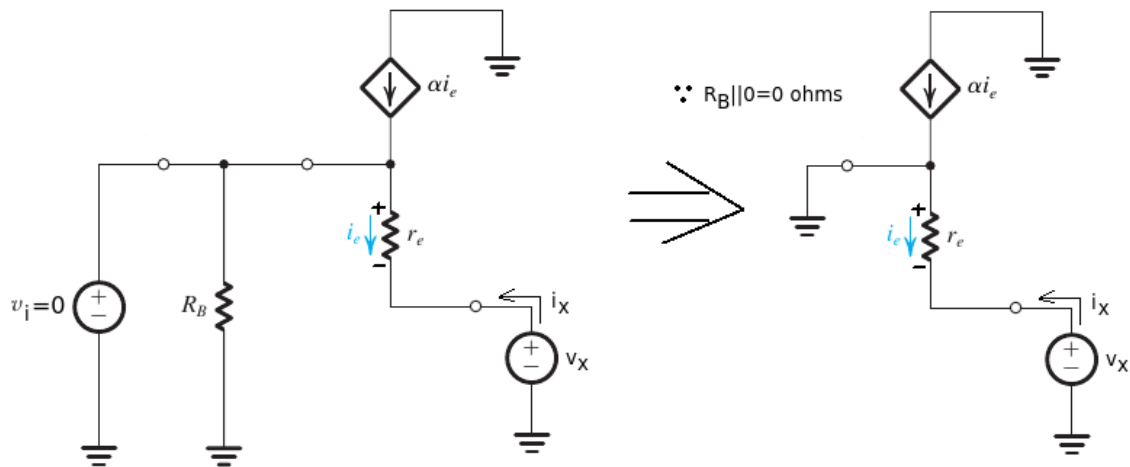
$R_o$

- the output resistance  $R_o$  is determined by removing  $R_L$  and looking into the emitter while setting  $v_i = 0$ 
  - here we set  $v_i = 0$



- setting  $v_i = 0 \Rightarrow$

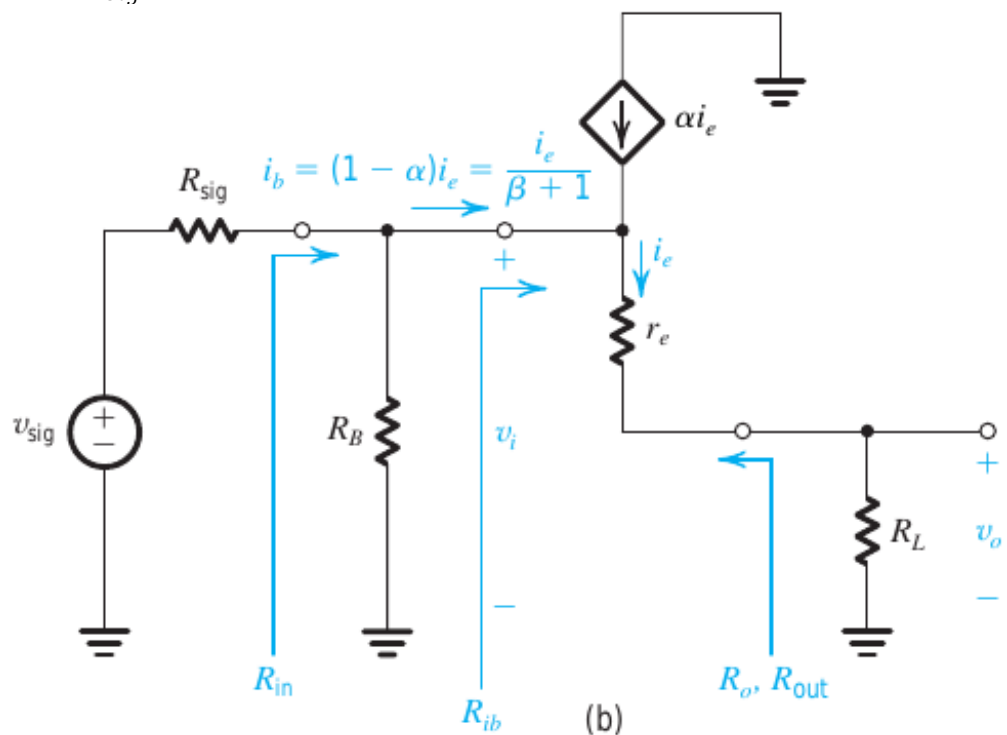




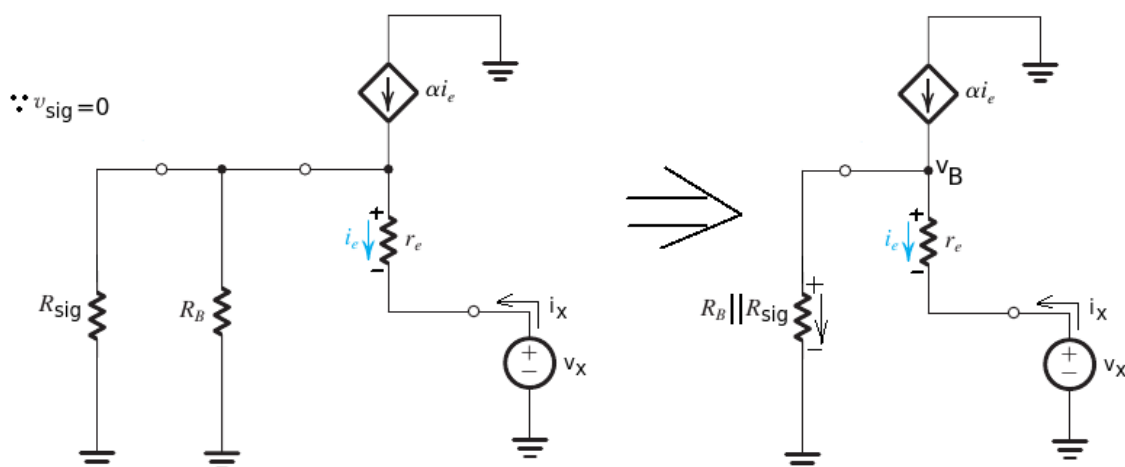
- ohms' law  $\Rightarrow i_e = \frac{0 - v_x}{r_e} = \frac{-v_x}{r_e}$
- $i_x = \frac{v_x}{r_e} \therefore i_e = -i_x$
- $\Rightarrow R_o = \frac{v_x}{i_x} = r_e$

$R_{out}$ :

- the output resistance  $R_{out}$  is determined by removing  $R_L$  and looking into the emitter while setting  $v_{sig} = 0$ 
  - here  $v_{sig} = 0$



- setting  $v_{sig} = 0$



○ KCL at  $v_B \Rightarrow \alpha i_e = i_e + \frac{v_B}{R_B \parallel R_{sig}}$

■  $\Rightarrow \alpha i_e - i_e = \frac{v_B}{R_B \parallel R_{sig}} \Rightarrow i_e(\alpha - 1) = \frac{v_B}{R_B \parallel R_{sig}}$

● KCL at  $v_B \Rightarrow \alpha i_e = i_e + \frac{v_B}{R_B \parallel R_{sig}}$

○  $\Rightarrow \alpha i_e - i_e = \frac{v_B}{R_B \parallel R_{sig}} \Rightarrow i_e(\alpha - 1) = \frac{v_B}{R_B \parallel R_{sig}}$

■  $i_e \left( \frac{\beta}{\beta+1} - 1 \right) = \frac{v_B}{R_B \parallel R_{sig}}$

■  $i_e \left( \frac{\beta - \beta - 1}{\beta+1} \right) = \frac{v_B}{R_B \parallel R_{sig}}$

■  $i_e \left( \frac{-1}{\beta+1} \right) = \frac{v_B}{R_B \parallel R_{sig}}$

■  $i_x \left( \frac{1}{\beta+1} \right) = \frac{v_B}{R_B \parallel R_{sig}} \because i_e = -i_x$

■  $i_x = \frac{(\beta+1)v_B}{R_B \parallel R_{sig}}$

■ by ohms law

■  $v_B - v_x = i_e r_e \Rightarrow v_B = v_x + i_e r_e = v_x - i_x r_e$

■  $\Rightarrow i_x = \frac{(\beta+1)v_B}{R_B \parallel R_{sig}} = \frac{(\beta+1)(v_x - i_x r_e)}{R_B \parallel R_{sig}}$

■  $\frac{i_x}{i_x} = \frac{(\beta+1)}{R_B \parallel R_{sig}} \left( \frac{v_x - i_x r_e}{i_x} \right) \because \text{divide both sides by } i_x$

■  $1 = \frac{(\beta+1)}{R_B \parallel R_{sig}} \left( \frac{v_x}{i_x} - \frac{i_x r_e}{i_x} \right) = \frac{(\beta+1)}{R_B \parallel R_{sig}} (R_{out} - r_e) \because R_{out} = \frac{v_x}{i_x}$

■  $\frac{(\beta+1)}{R_B \parallel R_{sig}} (R_{out} - r_e) = 1 \Rightarrow R_{out} - r_e = \frac{R_B \parallel R_{sig}}{(\beta+1)}$

■  $R_{out} = r_e + \frac{R_B \parallel R_{sig}}{(\beta+1)}$

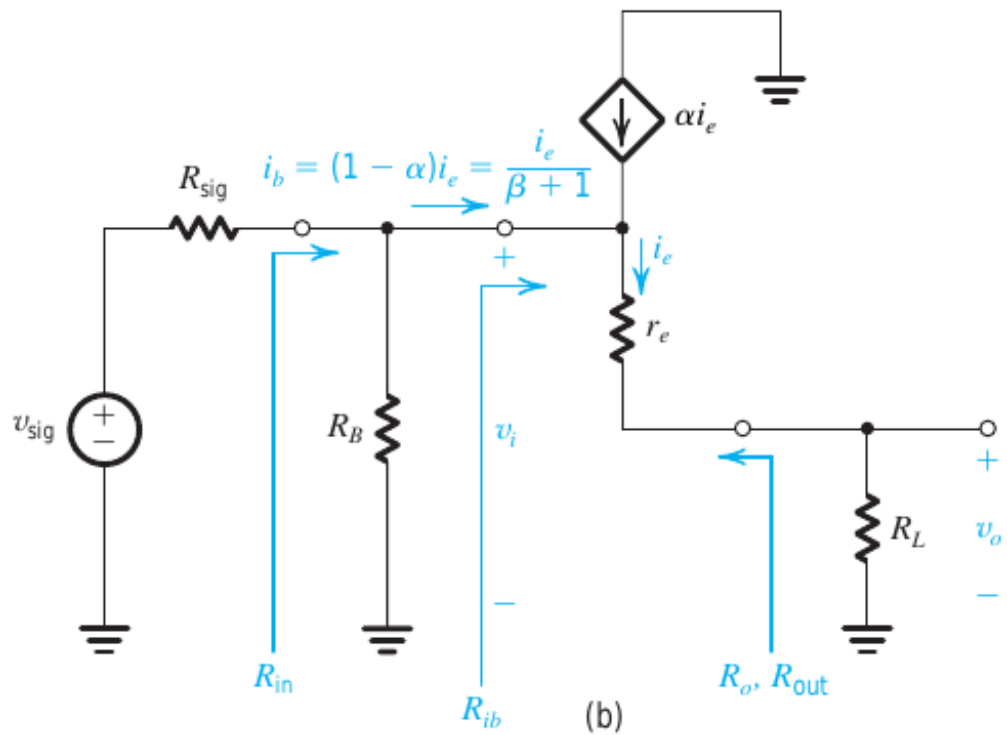
● The voltage gain  $A_v$  is given as

○  $A_v = \frac{v_o}{v_i}$

■ By voltage divider

■  $v_o = \frac{R_L}{R_L + r_e} v_i$

■ thus  $A_v = \frac{v_o}{v_i} = \frac{R_L}{r_e + R_L}$

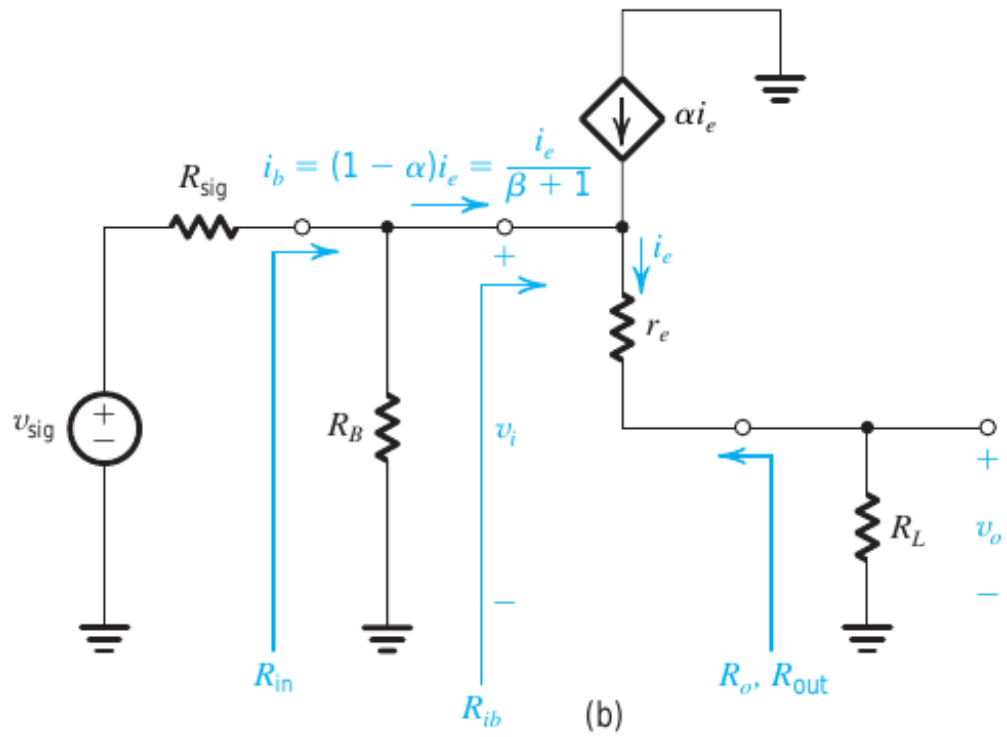


•  $A_v = \frac{v_o}{v_i} = \frac{R_L}{r_e + R_L}$

◦ and  $A_{vo}$  can be given as

■  $A_{vo} = \left. \frac{v_o}{v_i} \right|_{R_L = \infty} = \left. \frac{R_L}{r_e + R_L} \right|_{R_L = \infty}$

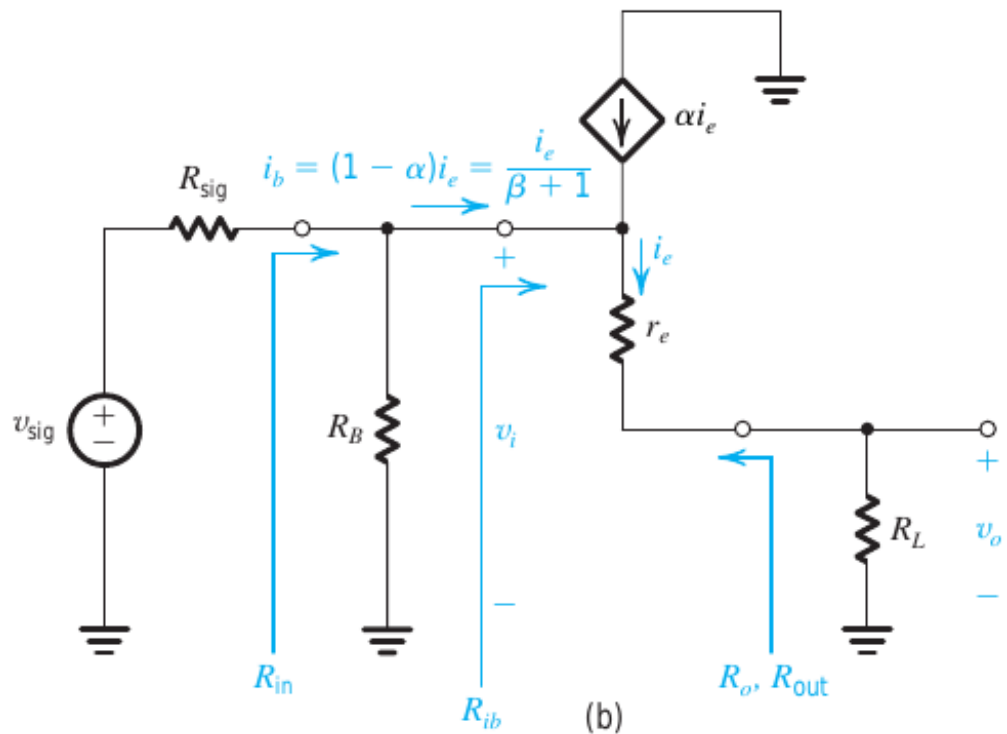
■  $A_{vo} = \left. \frac{1}{\frac{r_e}{R_L} + 1} \right|_{R_L = \infty} = \frac{1}{0+1} = 1$



- $G_v = \frac{v_o}{v_{sig}} = \frac{v_i}{v_{sig}} \frac{v_o}{v_i} = \frac{v_i}{v_{sig}} A_v$

- using voltage divider rule,  $v_i = \frac{R_{in}}{R_{in} + R_{sig}} v_{sig}$

- $G_v = \frac{v_i}{v_{sig}} A_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v$



- As  $R_{in} = R_B \parallel \{(\beta + 1)(r_e + R_L)\}$ ,  $A_v = \frac{R_L}{r_e + R_L}$

$$\begin{aligned}
& \circ G_v = \frac{R_{in}}{R_{in} + R_{sig}} A_v = \frac{R_B \parallel \{(\beta+1)(r_e + R_L)\}}{R_B \parallel \{(\beta+1)(r_e + R_L)\} + R_{sig}} \left( \frac{R_L}{r_e + R_L} \right) \\
\bullet G_v &= \frac{R_B \parallel \{(\beta+1)(r_e + R_L)\}}{R_B \parallel \{(\beta+1)(r_e + R_L)\} + R_{sig}} \left( \frac{R_L}{r_e + R_L} \right) \\
& \circ G_{vo} = .G_v|_{R_L \rightarrow \infty} = . \frac{R_B \parallel \{(\beta+1)(r_e + R_L)\}}{R_B \parallel \{(\beta+1)(r_e + R_L)\} + R_{sig}} \left( \frac{R_L}{r_e + R_L} \right) \Big|_{R_L \rightarrow \infty} \\
& \circ G_{vo} = . \frac{R_B \parallel \{(\beta+1)(r_e + R_L)\}}{R_B \parallel \{(\beta+1)(r_e + R_L)\} + R_{sig}} \left( \frac{1}{\frac{r_e}{R_L} + 1} \right) \Big|_{R_L \rightarrow \infty} \\
& \circ G_{vo} = \frac{R_B \parallel \{(\beta+1)(r_e + \infty)\}}{R_B \parallel \{(\beta+1)(r_e + \infty)\} + R_{sig}} \left( \frac{1}{\frac{r_e}{\infty} + 1} \right) \\
& \circ G_{vo} = \frac{R_B \parallel \infty}{(R_B \parallel \infty) + R_{sig}} \left( \frac{1}{0+1} \right) \\
& \circ G_{vo} = \frac{R_B}{R_B + R_{sig}}
\end{aligned}$$


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